

Parts F & G: Defect Log and Metrics

Cinema Distribution System

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Group / repository: Enterprise-F-Project / cinema-

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This document records defects discovered during testing and computes the course metrics: defect density, defect removal efficiency, defects found versus escaped, and coverage.

PART F - Defect management

F.1 Defect lifecycle used

Each defect is tracked as: New -> Assigned -> Fixed -> Verified -> Closed. Where a finding is accepted as documented product behaviour for the academic release, status Deferred is used (still visible in the log, not silently ignored).

Severity	Meaning
High	Blocks a core workflow or creates a serious security exposure
Medium	Incorrect but limited impact; workaround may exist
Low	Minor inconsistency; low user impact

Priority	Meaning
P1	Fix or decide immediately before demo/release recommendation
P2	Should be addressed in the next hardening cycle
P3	Nice to improve; acceptable residual risk if documented

F.2 Defect summary

ID	Title	Sev	Pri	Status
DEF-001	Public registration allows ADMIN role	High	P1	Deferred
DEF-002	Auth failures return HTTP 403 (not 401)	Medium	P2	Deferred
DEF-003	Disabled-account message not reached	Low	P3	Deferred
DEF-004	Rental UI date format rejected by API	High	P1	Closed

F.3 Detailed defect records

DEF-001: Public registration allows selecting ADMIN role

Severity: High
Priority: P1
Status: Deferred
Detected in: Phase 4 security integration tests
Steps: POST /api/auth/register with role=ADMIN and valid other fields.

Expected: Public self-registration must not grant ADMIN.
Actual: Registration succeeds with role ADMIN.
Lifecycle: New -> Assigned -> Deferred (documented finding)

DEF-002: Unauthenticated/invalid/expired/blacklisted JWT yields HTTP 403

Severity: Medium
Priority: P2
Status: Deferred
Detected in: Phase 4 security integration tests
Steps: GET /api/movies with no token or bad/expired/blacklisted Bearer token.
Expected: Prefer HTTP 401 for authentication failures.
Actual: Responses are HTTP 403 under current SecurityConfig.
Lifecycle: New -> Assigned -> Deferred

DEF-003: Disabled account login does not reach AuthService message

Severity: Low
Priority: P3
Status: Deferred
Detected in: Phase 4 disabled-account security test
Steps: Login with an inactive user (active=false) and correct password.
Expected: Clear Account is disabled message from AuthService.
Actual: Spring Security rejects authentication earlier.
Lifecycle: New -> Assigned -> Deferred

DEF-004: Rental create from UI fails: date-only vs LocalDateTime

Severity: High
Priority: P1
Status: Closed
Detected in: Phase 5 Selenium + API confirmation
Steps: Distributor creates movie+license; client submits rental using date inputs.
Expected: Rental created; sheet closes; status Requested.
Actual: HTTP 400 Invalid value for request field (UI sent YYYY-MM-DD).
Lifecycle: New -> Assigned -> Fixed (append T00:00:00) -> Verified -> Closed

Notes: DEF-001 to DEF-003 were not hidden by changing production code during security testing; they remain documented findings. DEF-004 blocked the rental E2E path and was fixed.

PART G - Metrics

Metrics use the defect log and JaCoCo from mvn clean test. Core business logic = AuthService, RentalService, MovieService, LicenseService.

G.1 Code coverage

Metric	Value	Interpretation
Core branch coverage	97.3% (71/73)	Exceeds instructor 90% and PDF 80% targets.
AuthService	100% (12/12)	Register/login/logout paths covered by unit tests.
MovieService	100% (12/12)	Create/update/ownership and listing scopes covered.
RentalService	96.8% (30/31)	Lifecycle and access rules well covered.
LicenseService	94.4% (17/18)	Create/access/findAll paths covered.
Overall project branches	~61%	Controllers/exceptions rely more on E2E than unit depth.

Interpretation: Focusing coverage on core services matches the course requirement. Uncovered project-wide branches are mainly peripheral packages.

G.2 Defect counts

Metric	Value	Interpretation
Defects found in testing	4	Three security findings + one UI/API contract defect.
Fixed before recommendation	1 (DEF-004)	Critical rental path repaired and re-verified.
Deferred / accepted risk	3	Documented residual risk for academic release.
Escaped defects (known)	0	No additional post-test escapes recorded.

G.3 Defect density

Defect density = defects found / size. Backend main Java is about 2.27 KLOC (line count under src/main/java).

Input	Value
Defects found (D)	4
Size (KLOC)	2.27
Defect density	$4 / 2.27 = 1.76$ defects per KLOC

Interpretation: On a small academic codebase density is sensitive to KLOC counting. The useful signal is that testing found a manageable set of real issues, including one blocking workflow defect that was removed.

G.4 Defect removal efficiency (DRE)

$DRE = F / (F + E)$, where F = found before release, E = escaped after release.

Input	Value
Found before release (F)	4
Escaped after testing (E)	0 known
DRE	$4 / (4 + 0) = 100\%$ (within this academic scope)

Interpretation: With no known escapes beyond the logged set, DRE is 100% for the project window. Deferred items are known residual risk, not escapes.

G.5 Found versus escaped

Category	Count	Notes
Found by security integration	3	DEF-001, DEF-002, DEF-003
Found by system testing	1	DEF-004 rental date contract
Escaped (unknown after testing)	0	None recorded after suites went green

Interpretation: Security integration and E2E validation caught issues that pure service unit tests would miss (security

configuration and UI-to-API contracts).

G.6 Test suite size (context)

Suite	Approx. count
Unit (services)	~61
Validation (controllers)	5
Security integration	19
Selenium system	3
Full local total	~88
CI suite (no Selenium)	85

Summary

- Four defects logged with lifecycle fields required by the course.
- One high-severity workflow defect closed after fix and retest.
- Three security findings deferred with explicit residual risk.
- Core branch coverage 97.3%; DRE 100% (known-escape scope); density 1.76 / KLOC.

End of Parts F and G - Defect Log and Metrics.